

MUDITHA PRIYASAD

Automation | Mechatronics | Mechanical Design Engineer

IESL Associate Member (AM-32795)

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PROFESSIONAL SUMMARY

Automation and Mechanical Engineer with a BSc (Hons) in Mechanical Engineering and hands-on experience designing, developing, testing and commissioning mechatronic systems, production machines and prototypes. Delivered 15+ engineering projects spanning PLC control, sensors, pneumatics, embedded/firmware integration, CAD, fabrication, troubleshooting and maintenance. Built low-cost systems that increased fan-base filling throughput 6x and eyelet-punching output 4x.

CORE SKILLS

- **Automation & industrial control:** PLC ladder logic, sensors, actuators, pneumatics, stepper motion, PID concepts, HMI/SCADA support, commissioning and fault finding.
- **Mechatronic design:** Mechanisms, machines, jigs and fixtures, 3D CAD, DfMA, component selection, fabrication support and technical drawings.
- **Embedded & software:** Arduino IDE, C++, Python, Raspberry Pi, Marlin and GRBL; firmware configuration, control logic and practical system integration.
- **Testing & delivery:** Prototype build, wiring awareness, system testing, preventive/corrective maintenance, performance improvement, installation support and procurement.
- **Tools:** SOLIDWORKS, AutoCAD, Siemens STEP 7 Micro/WIN, Delta ISPSOFT, Mitsubishi GX Works, Xinje PLC Software, Arduino IDE, MS Excel, MATLAB, OrcaSlicer and Cura.

PROFESSIONAL EXPERIENCE

MARQO Industries | Mechanical Design & Automation Engineer

Dec 2024 - Present

- Appointed as the company's first engineer; established the engineering and automation function, including workshop systems, tooling control, technical documentation and supporting software workflows.
- Designed and commissioned production machines, jigs, fixtures and pneumatic mechanisms from requirements and CAD through component selection, fabrication follow-up, testing, troubleshooting and operator feedback.
- Set up and troubleshoot spot-welding, packing, pad-printing and LED-cup installation equipment, resolving mechanical, pneumatic, alignment and process faults during production use.
- Delivered a semi-automatic fan-base machine (1 to 6 bases/min), a B22 eyelet puncher (3-4 to 16 pcs/min), a 1W LED-bulb assembly prototype (up to 20 pcs/min) and an ergonomic motor fixture.

MTone Designers | Founder

Jun 2021 - Present

- Delivered 20+ custom engineering and digital-manufacturing jobs, managing requirements, design, fabrication, assembly, inspection, costing and customer handover.
- Built, wired, calibrated and tested 3 custom FDM printers and 4 laser machines, integrating mechanical frames, electronics, firmware/control settings and safe operating procedures.

MAS Active Shadowline | Engineering Intern

Apr 2024 - Nov 2024

- Supported TPM, 5S, Kaizen, KPI tracking and operational checklists; prepared machine/process standards and a phased repair-workshop layout with utility and contractor coordination.

Effective Engineering & Automation | Automation Engineering Intern

Feb 2023 - Aug 2023

- Supported design, fabrication, commissioning, troubleshooting and maintenance for tensile-testing, concrete-mixer, oil-filling, glove-production and screen-printing equipment.
- Delivered a custom FDM 3D printer end-to-end: mechanical architecture, procurement, wiring, BTT SKR 1.4/Marlin configuration, calibration, fault correction and print validation.

Effective Engineering & Automation | Automation Engineering Intern (continued)

- Diagnosed Haiwell and Xinje PLC systems, checked relay/NPN/PNP inputs and expansion modules, tested ladder routines, completed three-phase control wiring and improved a Haiwell SCADA interface.

MAS Kreedha Vaanavil | Automation Engineering Intern

Jan 2019 - Apr 2019

- Integrated mechanical, pneumatic, electrical and control elements for a bonding-tape machine using PLC sequencing, stepper drive, PID temperature control, SSR, thermocouple, heaters, sensors and cylinders.
- Supported iterative testing for tape length, temperature stability, positioning, cylinder sensing, fabric grip and two-hand start protection, improving repeatability and operator safety.

SELECTED AUTOMATION & R&D PROJECTS

- **Custom FDM 3D printer:** Designed the mechanical architecture, managed procurement and wiring, configured BTT SKR 1.4 and Marlin, calibrated the machine and validated successful prints.
- **Fan-base filling machine:** Designed and commissioned a semi-automatic pneumatic filling and assembly cycle; increased output from 1 to 6 bases/min while reducing manual effort.
- **1W LED-bulb assembly machine:** Developed a semi-automatic prototype integrating sensors, pneumatics and control logic, with capacity up to 20 bulbs/min.
- **Bio-inspired hopping robot:** Designed and fabricated a two-actuator grasshopper-inspired jumping mechanism; worked on actuator selection, control and prototype tuning.
- **Line-follower robot with gripper:** Built sensor-guided autonomous movement and a liftable gripper for a simple pick-and-place task.
- **CNC PCB drilling machine:** Built compact stepper-driven mechanics and stacked positioning to process 18 PCB pieces per set-up with improved alignment and repeatability.

INDUSTRIAL MACHINE & PROCESS EXPOSURE

- **Production & assembly:** Spot welding, packing, pad printing, LED-cup installation, fan-base filling, eyelet punching and LED-bulb assembly.
- **Service & troubleshooting:** Tensile-testing, concrete-mixer, oil-filling, glove-production and screen-printing equipment.
- **Automation technologies:** PLC/HMI support, sensors, actuators, pneumatics, stepper motion, PID concepts, control logic and fault isolation.
- **Documentation & safety:** Technical drawings, SOPs, machine/process standards, project reports, operator guidance, TPM and safe-start controls.

EDUCATION

University of Jaffna | BSc (Hons) Mechanical Engineering

Oct 2017 - Feb 2024

- Final-year research: designed and fabricated the two-actuator bio-inspired hopping robot; led practical engineering projects and exhibition teams.

CERTIFICATIONS & TRAINING

- **SOLIDWORKS:** CSWA - Mechanical Design; CSWA-AM - Additive Manufacturing.
- **Automation:** PLC Training - University of Moratuwa (2025); PLC Programming - Epic Engineering (2024).
- **Maintenance & controls:** Maintenance of Industrial Plant & Machinery (2026) and Industrial Motor Operation & Control (2025) - NERDC.
- **Manufacturing:** Advanced Injection Moulding Technology - Industrial Development Board of Ceylon (2026).

ACHIEVEMENTS & PROFESSIONAL ACTIVITIES

- Bronze Award / 3rd place for the University of Jaffna engineering-project display at Techno Sri Lanka 2023; participant at Air Tattoo 2024 and SLT Robot Battles 2019.
- Organiser - University of Jaffna Battlegrounds 2020; IESL Associate Member AM-32795. | Languages: English - Professional proficiency; Sinhala - Native.